MEDPOWER 2026

15th Mediterranean Conference on Power Generation, Transmission, Distribution and Energy Conversion

03-06 November 2026 • Paphos, Cyprus

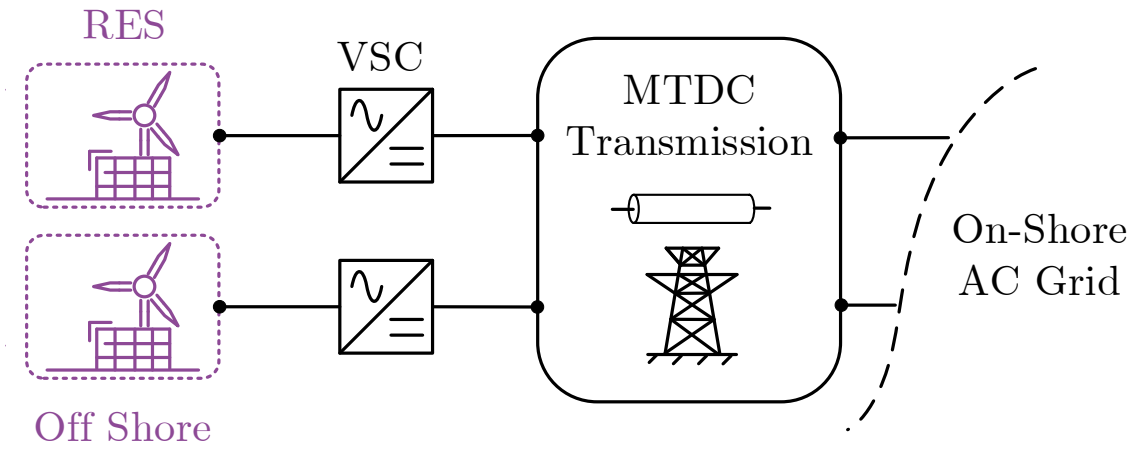
AC/DC OPF with Renewable Energy Source-Side Privacy Preservation: Distributed Method vs. Projection Method

Haixiao Li, Sunny Singh, Aleksandra Lekić*

Intelligent Electrical Power Grids (IEPG) Group,
Delft University of Technology, Delft, Netherlands

1. Motivation

AC/DC OPF with Renewable Energy Source-Side Privacy Preservation: Distributed Method vs. Projection Method



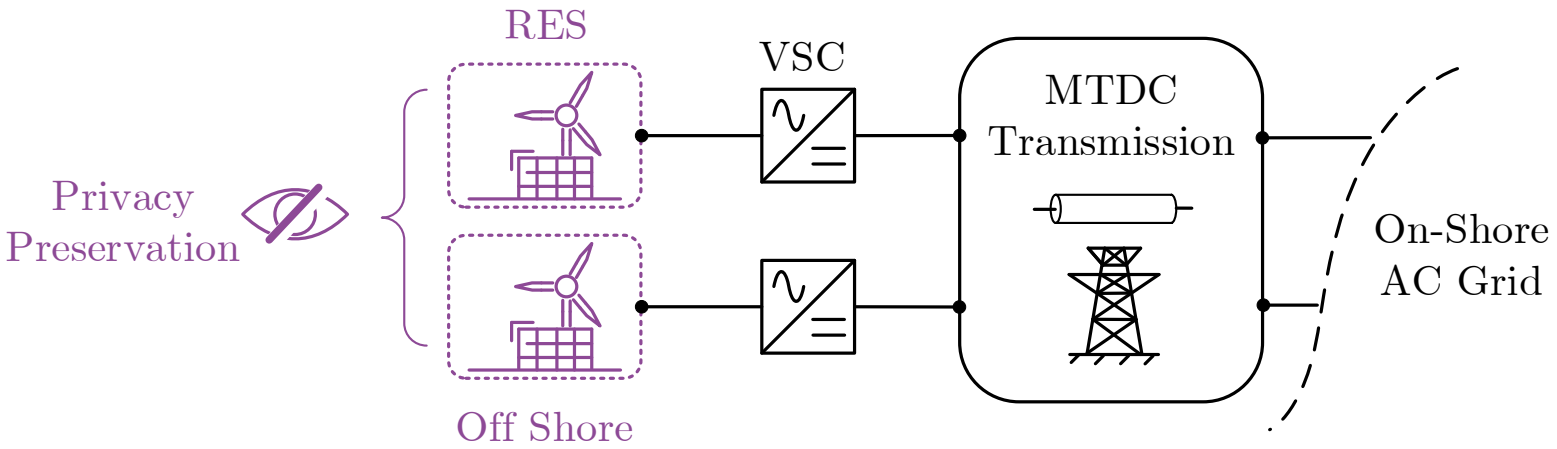
✎ OPF: Optimal Power Flow; VSC: Voltage Source Converter; RES: Renewable Energy Source; MTDC: Multi-Terminal DC; TSO: Transmission System Operator

Key Considerations

- Use AC/DC OPF to achieve optimal power dispatch(Gen, RES) for economical operation
- TSO achieve global optimal decision-making without access to RES internal parameters
- Distributed- and projection- based methods can handle RES privacy-preserving issue using limited boundary information to acquire global optimal solutions

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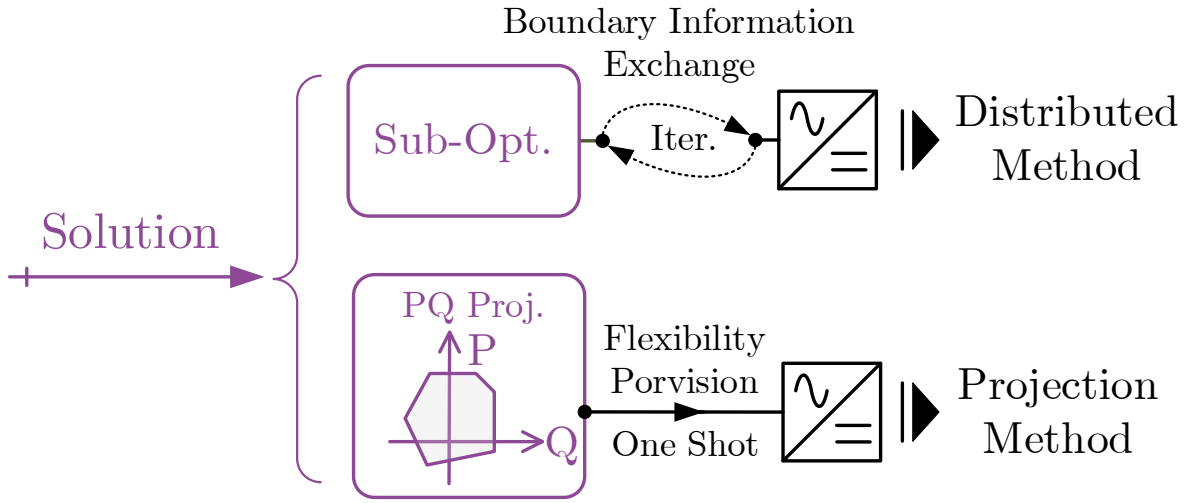
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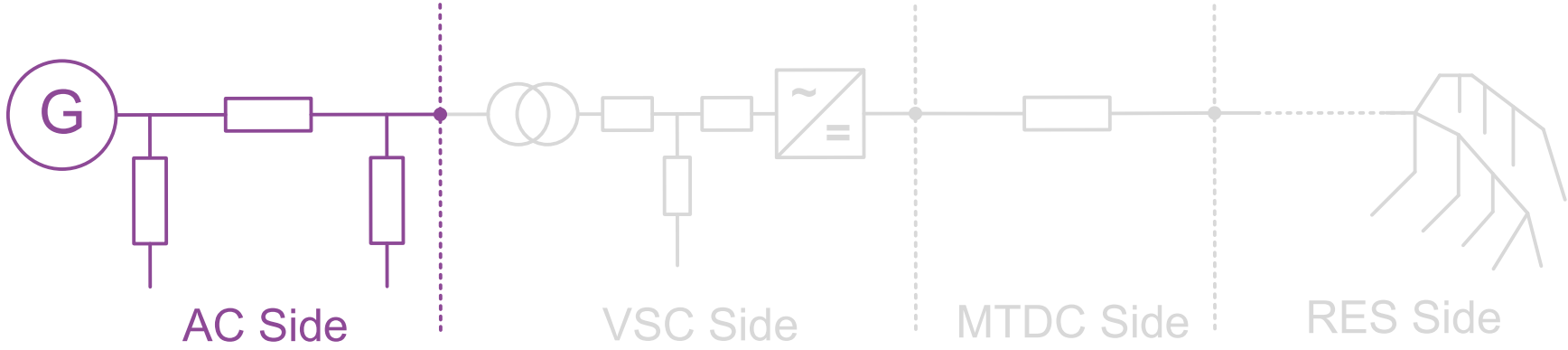


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2. Model Formulation –SOCP relaxed AC power flow



- SOCP: Second-Order Cone Programming

CONSTRAINTS OF THE AC GRID

AC branches

$$p_{ij}^{\mathrm{ac}} = g_{ij}^{\mathrm{ac}} \left(c_{ii}^{\mathrm{ac}} - c_{ij}^{\mathrm{ac}}\right) + b_{ij}^{\mathrm{ac}} s_{ij}^{\mathrm{ac}}, \quad q_{ij}^{\mathrm{ac}} = -b_{ij}^{\mathrm{ac}} \left(c_{ii}^{\mathrm{ac}} - c_{ij}^{\mathrm{ac}}\right) + g_{ij}^{\mathrm{ac}} s_{ij}^{\mathrm{ac}}, \qquad \forall (i,j) \in \mathcal{E}^{\mathrm{ac}}.$$

$$-\bar{s}_{ij}^{\mathrm{ac}} \leq \cos\left(\frac{k}{N}\pi\right) p_{ij}^{\mathrm{ac}} + \sin\left(\frac{k}{N}\pi\right) q_{ij}^{\mathrm{ac}} \leq \bar{s}_{ij}^{\mathrm{ac}}, \quad \forall k \in \{1, 2, \cdots, N\}, \quad \forall (i,j) \in \mathcal{E}^{\mathrm{ac}}.$$

AC buses

$$p_i^{\mathrm{ac}} = \sum_{j:(i,j)\in\mathcal{E}^{\mathrm{ac}}} p_{ij}^{\mathrm{ac}} + c_{ii}^{\mathrm{ac}} G_{i,\mathrm{sh}}^{\mathrm{ac}}, \quad q_i^{\mathrm{ac}} = \sum_{j:(i,j)\in\mathcal{E}^{\mathrm{ac}}} q_{ij}^{\mathrm{ac}} - c_{ii}^{\mathrm{ac}} B_{i,\mathrm{sh}}^{\mathrm{ac}}, \quad \forall i \in \mathcal{N}^{\mathrm{ac}}$$

$$p_i^{\mathrm{ac}} = p_{i,\mathrm{gen}}^{\mathrm{ac}} - p_{i,\mathrm{load}}^{\mathrm{ac}} - p_{i,\mathrm{a2v}}^{\mathrm{ac}}, \quad q_i^{\mathrm{ac}} = q_{i,\mathrm{gen}}^{\mathrm{ac}} - q_{i,\mathrm{load}}^{\mathrm{ac}} - q_{i,\mathrm{a2v}}^{\mathrm{ac}}, \quad \forall i \in \mathcal{N}^{\mathrm{ac}}.$$

$$(\underline{v}_i^{\mathrm{ac}})^2 \leq c_{ii}^{\mathrm{ac}} \leq (\bar{v}_i^{\mathrm{ac}})^2, \quad \forall i \in \mathcal{N}^{\mathrm{ac}}.$$

Generators

$$\underline{p}_{i,\mathrm{gen}}^{\mathrm{ac}} \leq p_{i,\mathrm{gen}}^{\mathrm{ac}} \leq \bar{p}_{i,\mathrm{gen}}^{\mathrm{ac}}, \quad \underline{q}_{i,\mathrm{gen}}^{\mathrm{ac}} \leq q_{i,\mathrm{gen}}^{\mathrm{ac}} \leq \bar{q}_{i,\mathrm{gen}}^{\mathrm{ac}}, \quad \forall i \in \mathcal{N}^{\mathrm{ac}}.$$

SOCP relaxation

$$\left(c_{ij}^{\mathrm{ac}}\right)^2 + \left(s_{ij}^{\mathrm{ac}}\right)^2 \leq c_{ii}^{\mathrm{ac}} c_{jj}^{\mathrm{ac}}, \quad \forall i \in \mathcal{N}^{\mathrm{ac}}, \quad \forall (i,j) \in \mathcal{E}^{\mathrm{ac}}.$$

(7)

(1) Branch power flow

(2) Branch capacity constraint

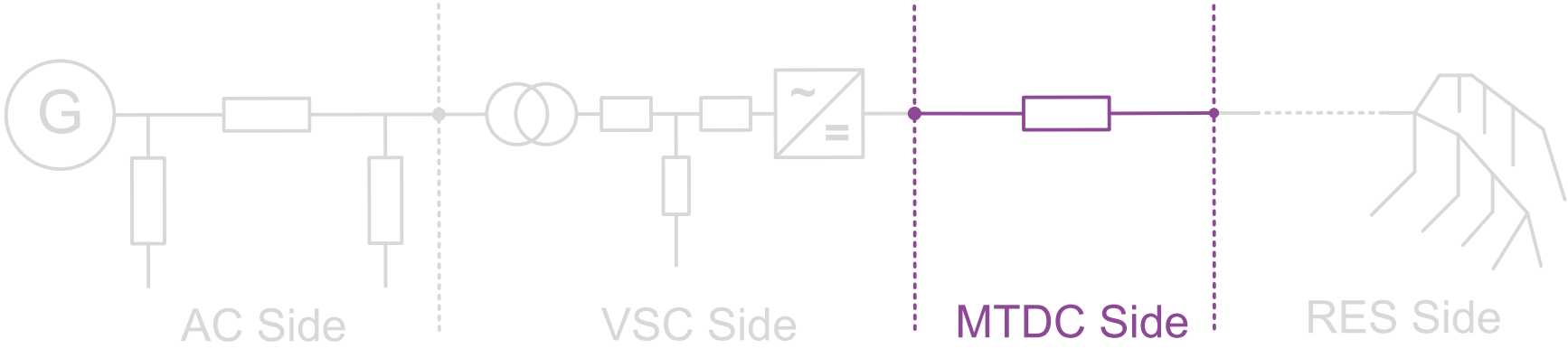
(3) Bus power injection

(4) Bus power balance

(5) Bus voltage bounds

(6) Generator power output limits

2. Model Formulation –SOCP relaxed DC power flow



- SOCP: Second-Order Cone Programming

CONSTRAINTS OF THE MTDC GRID

DC branches

$$p_{jk}^{\mathrm{dc}} + p_{kj}^{\mathrm{dc}} = r_{jk}^{\mathrm{dc}} l_{jk}^{\mathrm{dc}}, \quad \forall (j,k) \in \mathcal{E}^{\mathrm{dc}}. \tag{8}$$

$$u_j^{\mathrm{dc}} - u_k^{\mathrm{dc}} = r_{jk}^{\mathrm{dc}} \left(p_{jk}^{\mathrm{dc}} - p_{kj}^{\mathrm{dc}} \right), \quad \forall j,k \in \mathcal{N}^{\mathrm{dc}}, \quad \forall (j,k) \in \mathcal{E}^{\mathrm{dc}}. \tag{9}$$

Branch power flow

$$\underline{p}_{jk}^{\mathrm{dc}} \le p_{jk}^{\mathrm{dc}} \le \bar{p}_{jk}^{\mathrm{dc}}, \quad \forall (j,k) \in \mathcal{E}^{\mathrm{dc}}. \tag{10}$$

Branch capacity limit

DC buses

$$p_j^{\mathrm{dc}} = \rho^{\mathrm{dc}} \sum_{k:(j,k)\in\mathcal{E}^{\mathrm{dc}}} p_{jk}^{\mathrm{dc}}, \quad \forall j \in \mathcal{N}^{\mathrm{dc}}. \tag{11}$$

Bus power balance

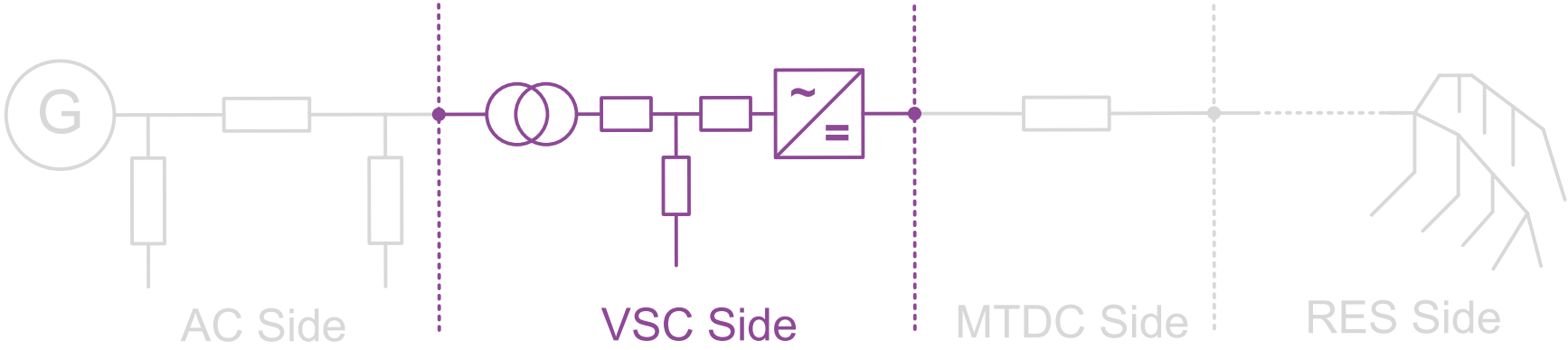
$$\left(\underline{v}_j^{\mathrm{dc}}\right)^2 \le u_j^{\mathrm{dc}} \le \left(\bar{v}_j^{\mathrm{dc}}\right)^2, \quad \forall j \in \mathcal{N}^{\mathrm{dc}}. \tag{12}$$

Bus voltage bounds

SOCP relaxation

$$\left(p_{jk}^{\mathrm{dc}}\right)^2 \le l_{jk}^{\mathrm{dc}} u_j^{\mathrm{dc}}, \quad \forall j \in \mathcal{N}^{\mathrm{dc}}, \quad \forall (j,k) \in \mathcal{E}^{\mathrm{dc}}. \tag{13}$$

2. Model Formulation –SOCP relaxed VSC power flow



- SOCP: Second-Order Cone Programming

CONSTRAINTS OF THE VSC STATION#v

VSC buses

$$p_s^{vsc,v} = c_{ss}^{vsc,v} g_{sc}^{vsc,v} - c_{sc}^{vsc,v} g_{sc}^{vsc,v} + s_{sc}^{vsc,v} b_{sc}^{vsc,v}, \quad q_s^{vsc,v} = -c_{ss}^{vsc,v} b_{sc}^{vsc,v} + c_{sc}^{vsc,v} b_{sc}^{vsc,v} + s_{sc}^{vsc,v} g_{sc}^{vsc,v}, \quad s \in \mathcal{N}^{vsc,v}, \quad (s,c) \in \mathcal{E}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{16}$$

$$p_c^{vsc,v} = c_{cc}^{vsc,v} g_{cs}^{vsc,v} - c_{cs}^{vsc,v} g_{cs}^{vsc,v} + s_{cs}^{vsc,v} b_{cs}^{vsc,v}, \quad q_c^{vsc,v} = -c_{cc}^{vsc,v} b_{cs}^{vsc,v} + c_{cs}^{vsc,v} b_{cs}^{vsc,v} + s_{cs}^{vsc,v} g_{cs}^{vsc,v}, \quad s,c \in \mathcal{N}^{vsc,v}, \quad (s,c) \in \mathcal{E}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{17}$$

Bus power balance in the AC side

$$(\underline{v}_s^{vsc,v})^2 \le c_{ss}^{vsc,v} \le (\bar{v}_s^{vsc,v})^2, \quad (\underline{v}_c^{vsc,v})^2 \le c_{cc}^{vsc,v} \le (\bar{v}_c^{vsc,v})^2, \quad s,c \in \mathcal{N}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{18}$$

Bus voltage bounds in the AC side

VSC power loss

$$p_c^{vsc,v} + p_{loss}^{vsc,v} + p_j^{dc} = 0, \quad \forall (j,v) \in \mathcal{A}^{d2v}, \quad c \in \mathcal{N}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{19}$$

Power coupling between AC/DC terminals

$$p_{loss}^{vsc,v} = a_{loss}^{vsc,v} + b_{loss}^{vsc,v} I_c^{vsc,v} + c_{loss}^{vsc,v} l_c^{vsc,v}, \quad c \in \mathcal{N}^{vsc,v}. \tag{20}$$

Converter loss

$$0 \le I_c^{vsc,v} \le \bar{I}_c^{vsc,v}, \quad 0 \le l_c^{vsc,v} \le (\bar{I}_c^{vsc,v})^2, \quad c \in \mathcal{N}^{vsc,v}. \tag{21}$$

Converter current limit

Coupling relationships

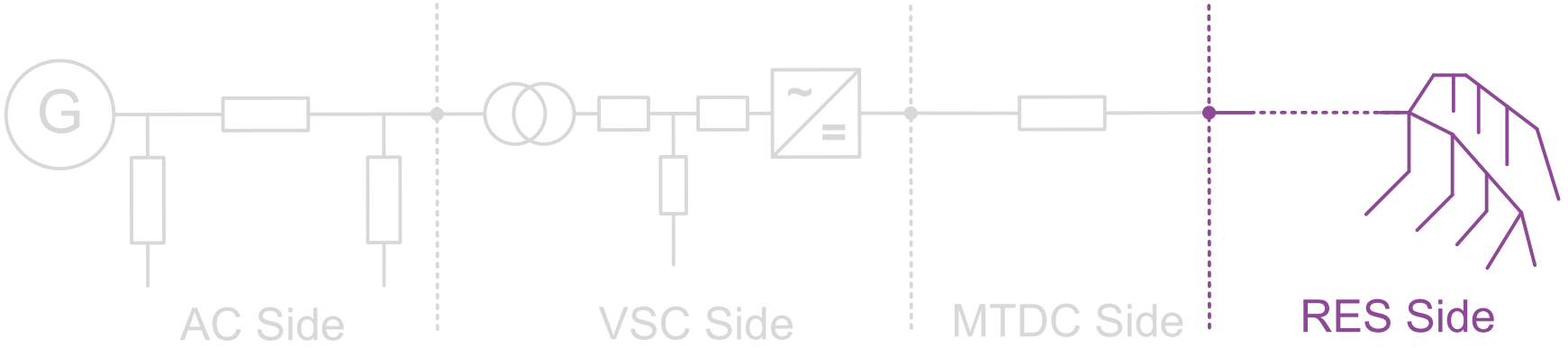
$$\begin{cases} c_{ss}^{vsc,v} = c_{ii}^{ac}, \quad p_s^{vsc,v} = p_{i,a2v}^{ac}, \quad q_s^{vsc,v} = q_{i,a2v}^{ac}, & \forall (i,v) \in \mathcal{A}^{a2v}, \\ c_{ss}^{vsc,v} = u_0^{res,r}, \quad p_s^{vsc,v} = p_{r2v}^{res,r}, \quad q_s^{vsc,v} = q_{r2v}^{res,r}, & \forall (r,v) \in \mathcal{A}^{r2v}, \end{cases} \quad s \in \mathcal{N}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{22}$$

Power coupling at the AC terminal

SOCP relaxation

$$(c_{sc}^{vsc,v})^2 + (s_{sc}^{vsc,v})^2 \le c_{ss}^{vsc,v} c_{cc}^{vsc,v}, \quad (I_c^{vsc,v})^2 \le l_c^{vsc,v}, \quad (p_c^{vsc,v})^2 + (q_c^{vsc,v})^2 \le c_{cc}^{vsc,v} l_c^{vsc,v}, \quad s,c \in \mathcal{N}^{vsc,v}, \quad (s,c) \in \mathcal{E}^{vsc,v}, \quad \forall v \in \mathcal{V}. \tag{23}$$

2. Model Formulation –Linearzied RES power flow



CONSTRAINTS OF RES PLANT #r

RES branches

$$p_{ij}^{\mathrm{res},r} = \sum_{k\in\mathcal{C}_r(j)} p_{jk}^{\mathrm{res},r} - p_j^{\mathrm{res},r}, \quad q_{ij}^{\mathrm{res},r} = \sum_{k\in\mathcal{C}_r(j)} q_{jk}^{\mathrm{res},r} - q_j^{\mathrm{res},r},$$
$$\forall j \in \mathcal{N}^{\mathrm{res},r}\setminus\{0\}, \quad i=\pi_r(j), \quad \forall r\in\mathcal{R}.$$
$$u_j^{\mathrm{res},r} = u_i^{\mathrm{res},r} - 2\left(r_{ij}^{\mathrm{res},r}p_{ij}^{\mathrm{res},r} + x_{ij}^{\mathrm{res},r}q_{ij}^{\mathrm{res},r}\right), \quad \forall (i,j)\in\mathcal{E}^{\mathrm{res},r}, \quad \forall r\in\mathcal{R}.$$

RES buses and PCC

$$p_i^{\mathrm{res},r} = p_{i,\mathrm{gen}}^{\mathrm{res},r}, \quad q_i^{\mathrm{res},r} = q_{i,\mathrm{gen}}^{\mathrm{res},r}, \quad \forall i\in\mathcal{N}^{\mathrm{res},r}\setminus\{0\}, \quad \forall r\in\mathcal{R}.$$
$$p_0^{\mathrm{res},r} = \sum_{k\in\mathcal{C}_r(0)} p_{0k}^{\mathrm{res},r} = -p_{\mathrm{r2v}}^{\mathrm{res},r},$$
$$q_0^{\mathrm{res},r} = \sum_{k\in\mathcal{C}_r(0)} q_{0k}^{\mathrm{res},r} = -q_{\mathrm{r2v}}^{\mathrm{res},r}, \quad \forall r\in\mathcal{R}.$$
$$\left(\underline{v}_i^{\mathrm{res},r}\right)^2 \le u_i^{\mathrm{res},r} \le \left(\bar{v}_i^{\mathrm{res},r}\right)^2, \quad \forall i\in\mathcal{N}^{\mathrm{res},r}, \quad \forall r\in\mathcal{R}.$$

RES units

$$0 \le p_{i,\mathrm{gen}}^{\mathrm{res},r} \le \bar{p}_{i,\mathrm{gen}}^{\mathrm{res},r},$$
$$-\bar{s}_{i,\mathrm{gen}}^{\mathrm{res},r} \le \cos\left(\frac{k\pi}{N}\right)p_{i,\mathrm{gen}}^{\mathrm{res},r} + \sin\left(\frac{k\pi}{N}\right)q_{i,\mathrm{gen}}^{\mathrm{res},r} \le \bar{s}_{i,\mathrm{gen}}^{\mathrm{res},r},$$
$$\forall k\in\{1,2,\ldots,N\}, \quad \forall i\in\mathcal{N}^{\mathrm{res},r}\setminus\{0\}, \quad \forall r\in\mathcal{R}.$$

(14), (15) — Lindistflow power model (for RES radial network)

(16) — Power injection on RES feeders

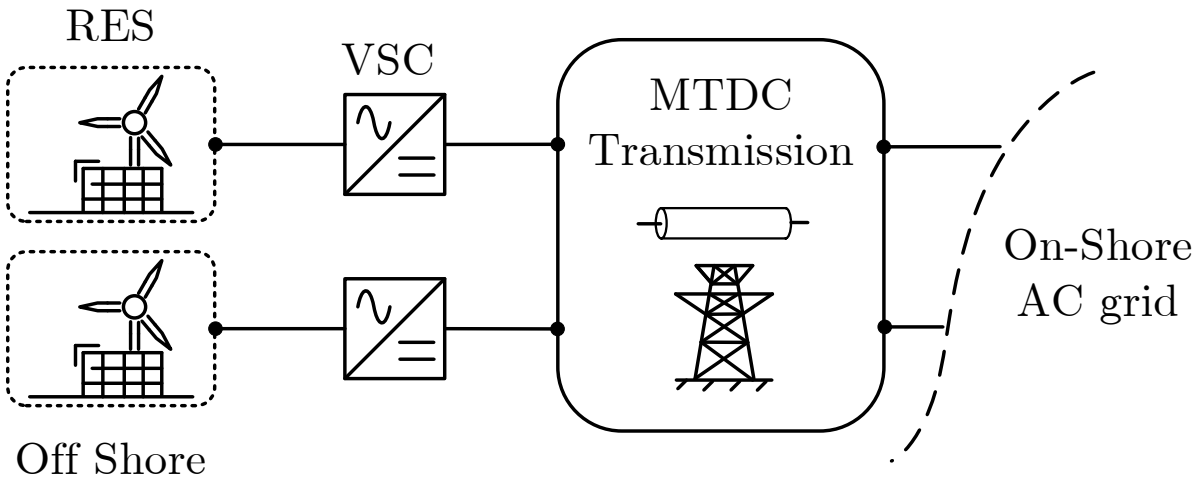
(17) — Power injection at PCC bus

(18) — Bus voltage limits

(19) — Power output limits of single RES unit

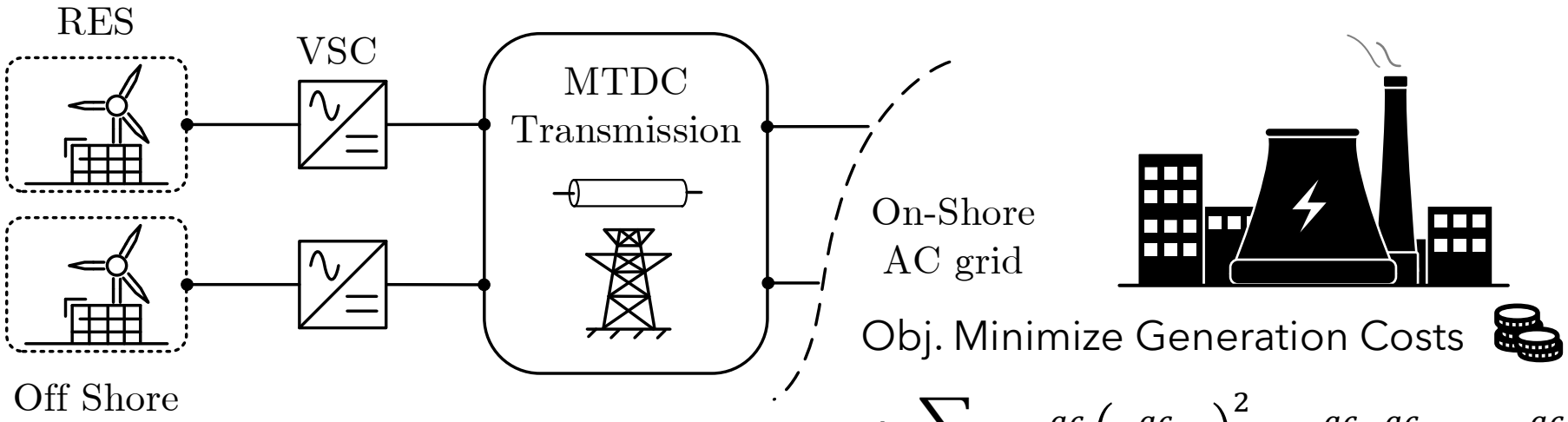
3. Methodology –Centralized Way

[Original Centralized Problem]



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[Original Centralized Problem]

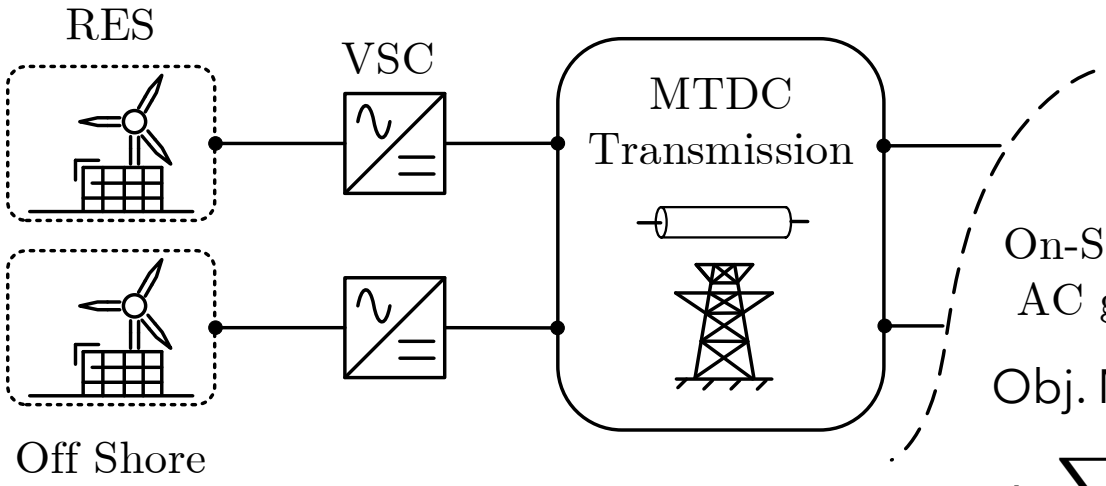


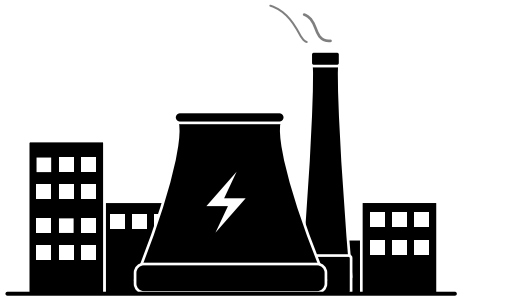
Obj. Minimize Generation Costs

$$\min \sum_{i \in \mathcal{N}^{ac}} c_{i,2}^{ac} \left(p_{i,gen}^{ac}\right)^2 + c_{i,1}^{ac} p_{i,gen}^{ac} + c_{i,0}^{ac}$$

3. Methodology –Centralized Way

[Original Centralized Problem]





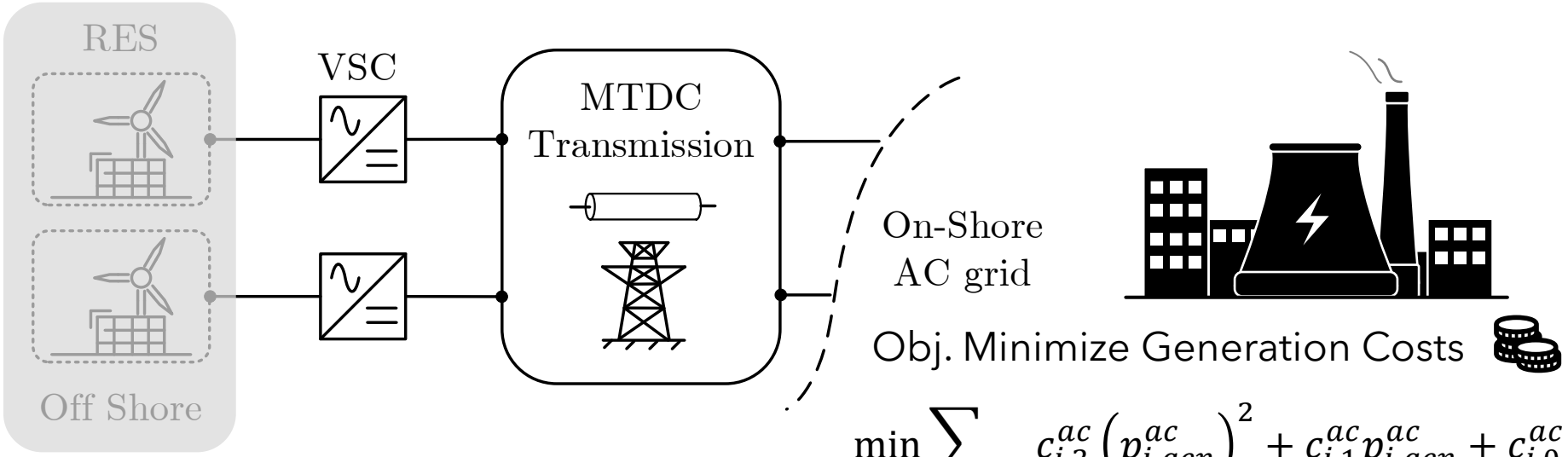
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RES-Side Constraint + VSC-Side Constraint + MTDC-Side Constraint + AC-Side Constraint

3. Methodology –Centralized Way

[Original Centralized Problem]



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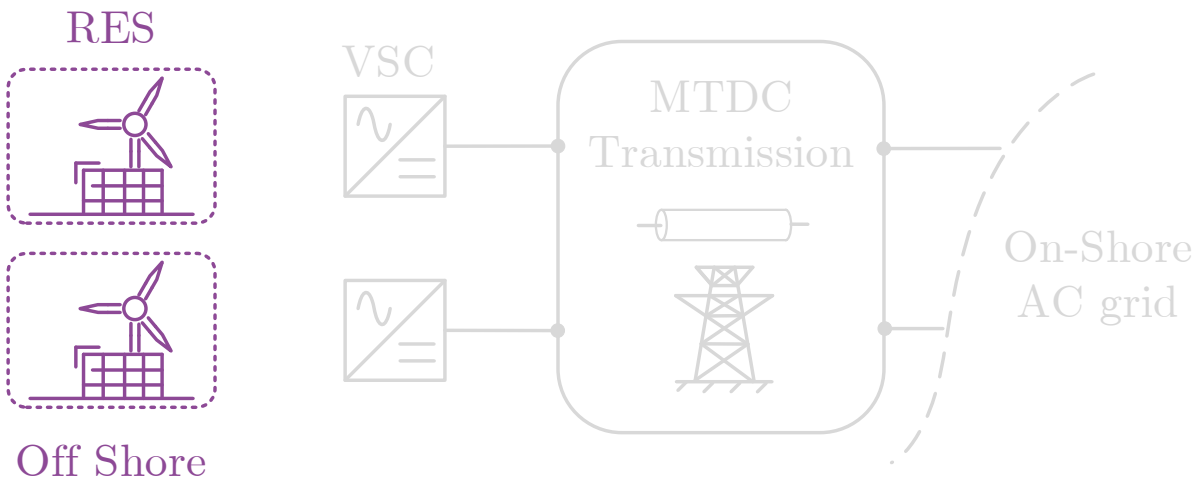
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RES-Side Constraint + VSC-Side Constraint + MTDC-Side Constraint + AC-Side Constraint

⚠ Due to the RES-side privacy-preserving issue, TSO cannot acquire RES-side constraint. Therefore, the problem cannot be solved in centralized way (。•́‿•̀。)

3. Methodology –Distributed Method

[RES-Side Sub-Problem] [TSO-Side Sub-Problem]



$$\text{Min} - \hat{\boldsymbol{\lambda}}^{[k]\top}\boldsymbol{Z} + \frac{\rho}{2}\left\|\hat{\boldsymbol{x}}^{[k]} - \boldsymbol{z}\right\|_2^2$$

S.t. RES-Side Constraint

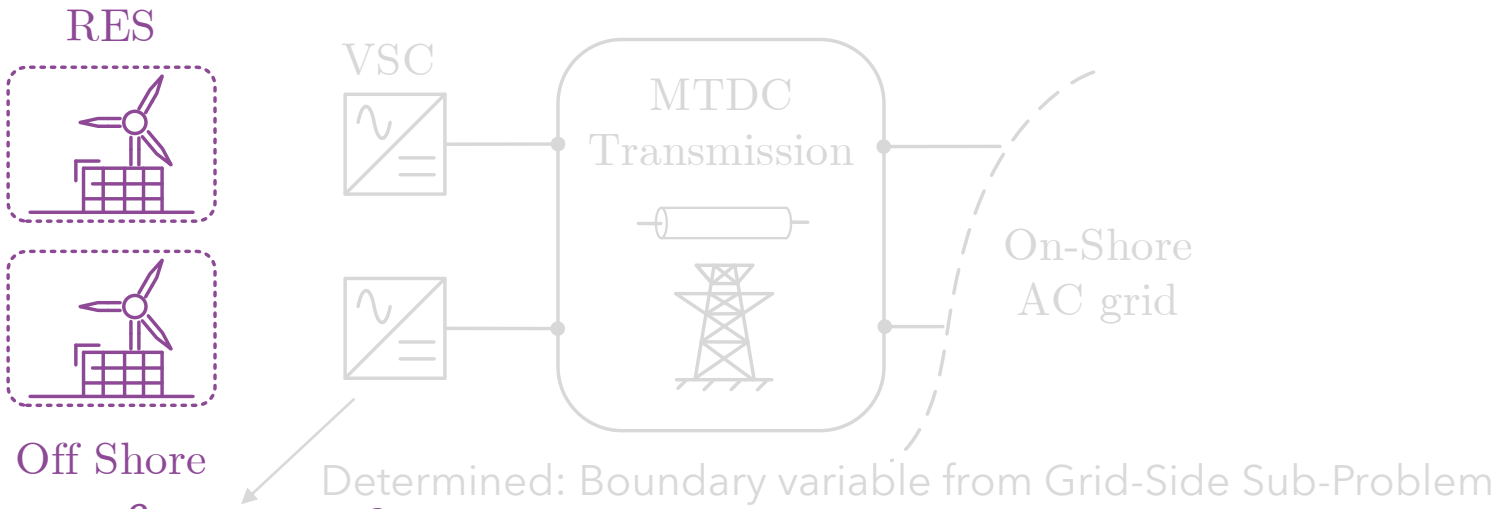
Key Steps of ADMM

- Solve the augmented Lagrangian function of the RES-Side Sub-Problem
- Solve the augmented Lagrangian function of the TSO-Side Sub-Problem
- Update the dual variables

ADMM: Alternating Direction Method of Multipliers

3. Methodology –Distributed Method

[RES-Side Sub-Problem] [TSO-Side Sub-Problem]



$$\text{Min} - \hat{\boldsymbol{\lambda}}^{[k]\top} \boldsymbol{Z} + \frac{\rho}{2} \left\| \hat{\boldsymbol{x}}^{[k]} - \boldsymbol{z} \right\|_2^2$$

S.t. RES-Side Constraint

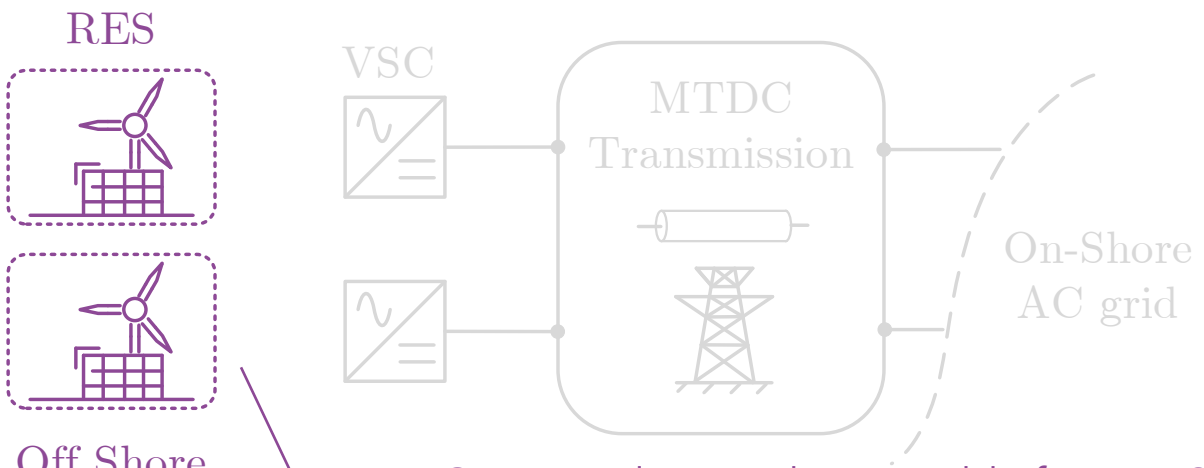
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✎ ADMM: Alternating Direction Method of Multipliers

3. Methodology –Distributed Method

[RES-Side Sub-Problem] [TSO-Side Sub-Problem]



To Be Optimized: Boundary variable from RES-Side Sub-Problem

$$\text{Min} - \hat{\boldsymbol{\lambda}}^{[k]\top} \boldsymbol{z} + \frac{\rho}{2} \left\| \hat{\boldsymbol{x}}^{[k]} - \boldsymbol{z} \right\|_2^2$$

S.t. RES-Side Constraint

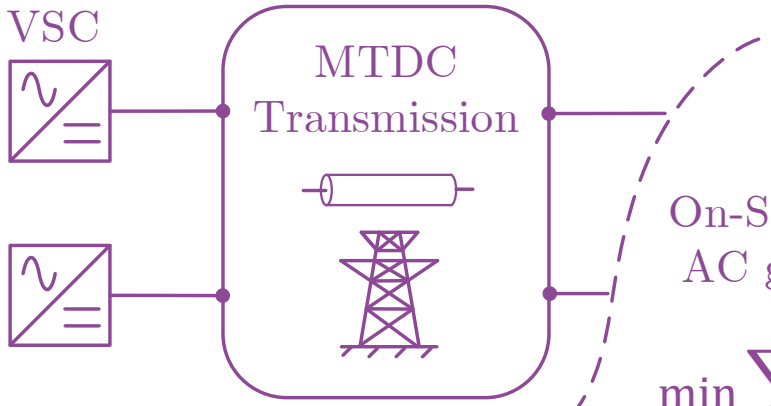
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✎ ADMM: Alternating Direction Method of Multipliers

3. Methodology –Distributed Method

[RES-Side Sub-Problem] [TSO-Side Sub-Problem]



S.t. VSC-Side Constraint +
MTDC-Side Constraint +
AC-Side Constraint

$$\min \sum_{i \in \mathcal{N}^{ac}} c_{i,2}^{ac} \left(p_{i,gen}^{ac}\right)^2 + c_{i,1}^{ac} p_{i,gen}^{ac} + c_{i,0}^{ac}$$

$$+\hat{\boldsymbol{\lambda}}^{[k]\top} \boldsymbol{x} + \frac{\rho}{2} \left\| \boldsymbol{x} - \hat{\boldsymbol{z}}^{[k]} \right\|_2^2$$

Key Steps of ADMM

- Solve the augmented Lagrangian function of the RES-Side Sub-Problem
- Solve the augmented Lagrangian function of the TSO-Side Sub-Problem
- Update the dual variables

✎ ADMM: Alternating Direction Method of Multipliers

3. Methodology –Distributed Method

$$\hat{\boldsymbol{\lambda}}^{[k+1]} \leftarrow \hat{\boldsymbol{\lambda}}^{[k]} + \rho\left(\hat{\boldsymbol{x}}^{[k]} - \hat{\boldsymbol{z}}^{[k]}\right)$$

Key Steps of ADMM

- Solve the augmented Lagrangian function of the RES-Side Sub-Problem
- Solve the augmented Lagrangian function of the TSO-Side Sub-Problem
- Update the dual variables

✎ ADMM: Alternating Direction Method of Multipliers

3. Methodology –Distributed Method

⏳ Check if residuals fall below the thresholds

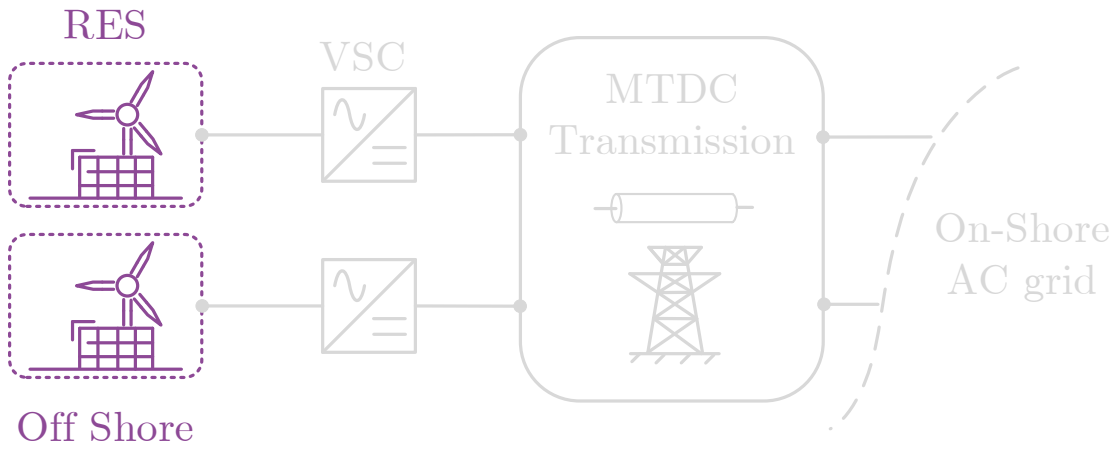
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✎ ADMM: Alternating Direction Method of Multipliers

3. Methodology –Projection Method

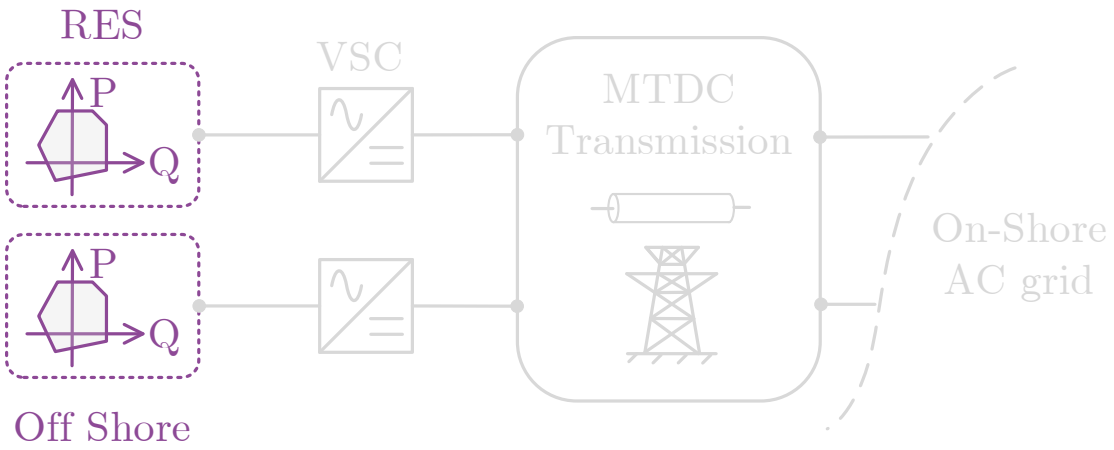
[RES-Side Optimization] [TSO-Side Optimization]



Internal constraints from RES-Side is unavailable, thus TSO-side optimization cannot be implemented (。•́ ⌓ •̀。)

3. Methodology –Projection Method

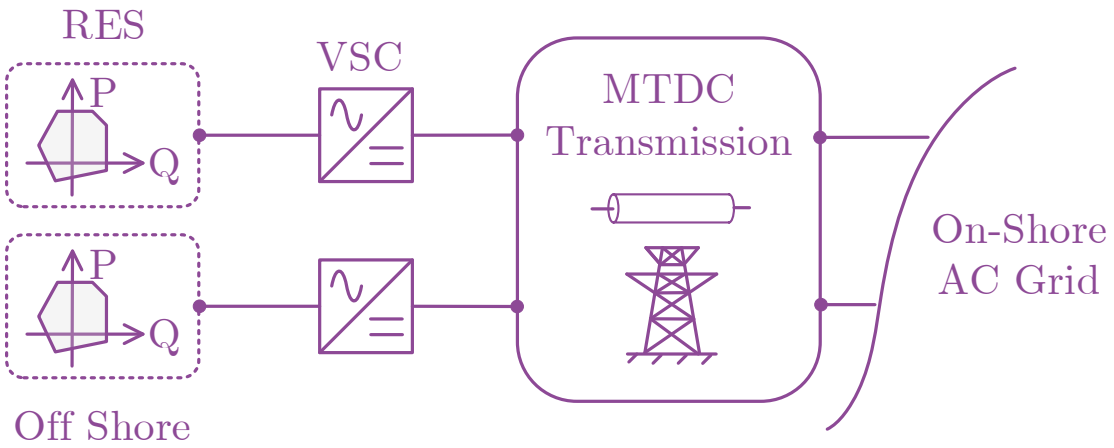
[RES-Side PQ Projection] [TSO-Side Optimization]



If we know PQ projection at the boundary(PCC) between RES- and TSO-side, then we can implement TSO-side optimization and achieve global optimum („• ᴗ •„)

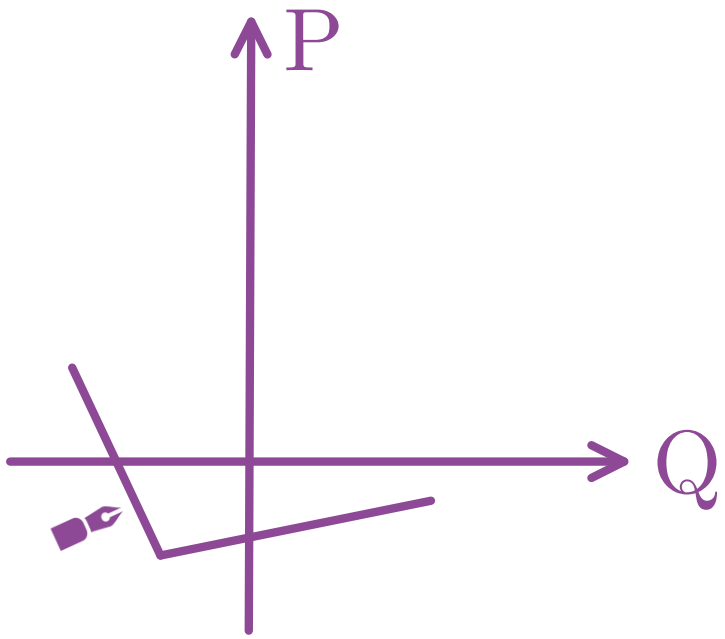
3. Methodology –Projection Method

[RES-Side PQ Projection] [TSO-Side Optimization]

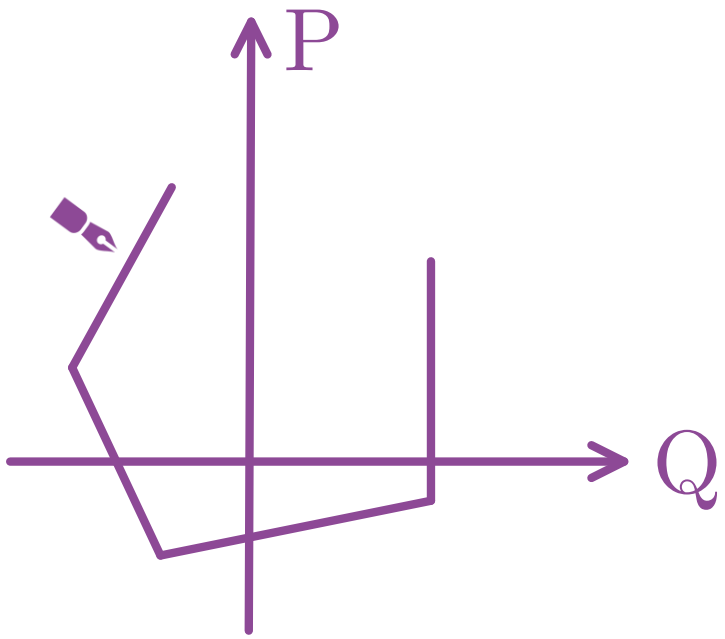


If we know PQ projection at the boundary(PCC) between RES- and TSO-side, then we can implement TSO-side optimization and achieve global optimum (ᵔᴥᵔ)

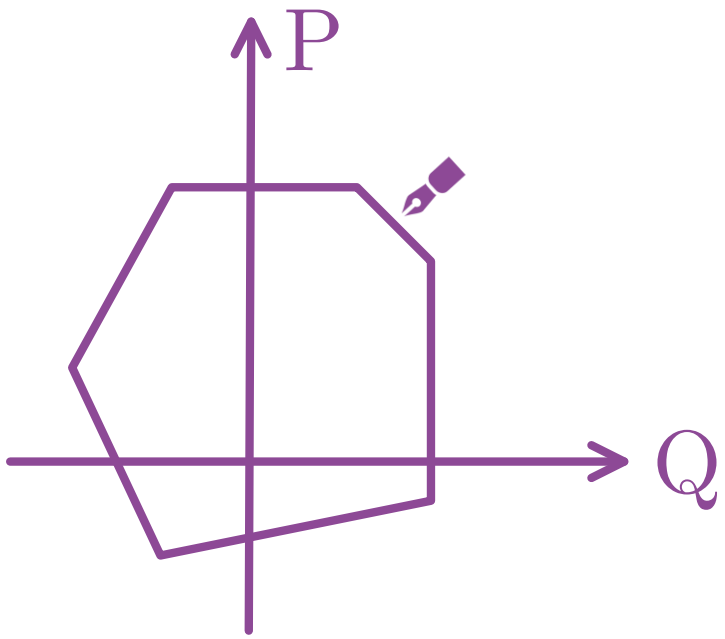
3. Methodology –Projection Method



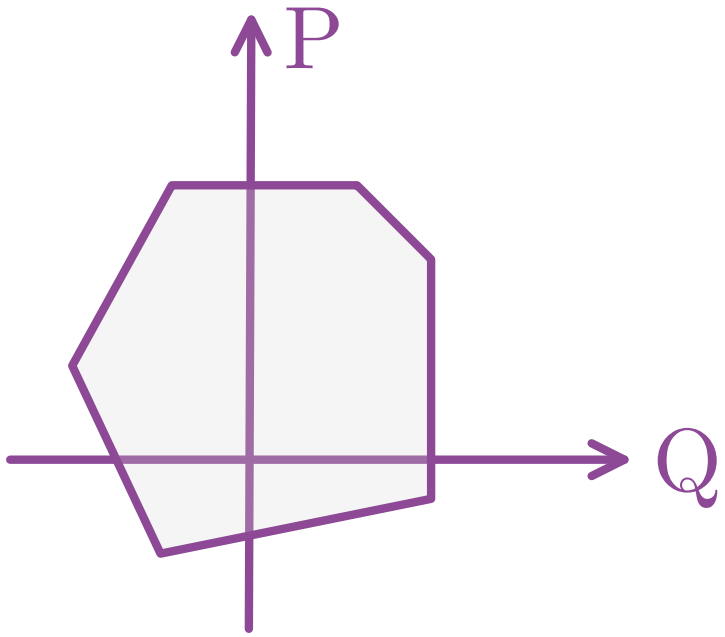
3. Methodology –Projection Method



3. Methodology –Projection Method

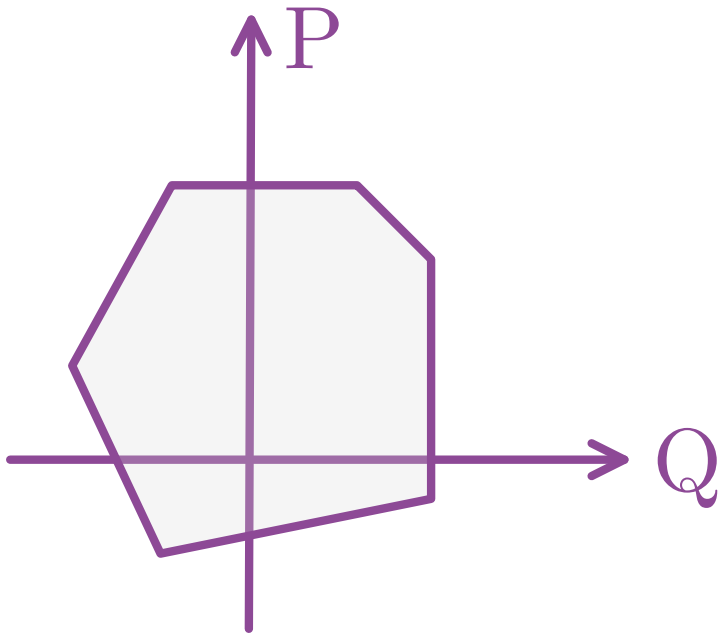


3. Methodology –Projection Method



How to acquire the projected PQ feasible region?

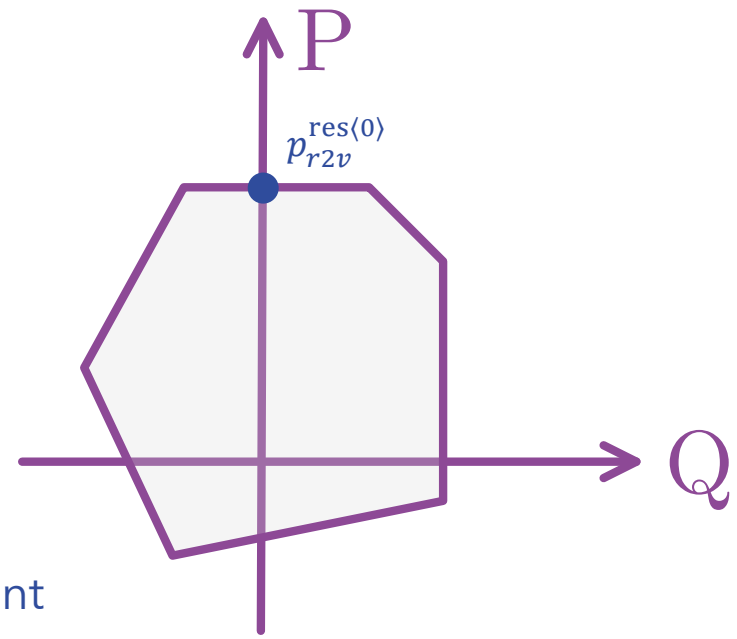
3. Methodology –Projection Method



$$\max/\min\{p_{r2v}^{res}, q_{r2v}^{res}\}$$

▶ The boundary of the projected PQ feasible region can be traced by solving a **multi-objective** optimization problem and identifying the **Pareto front** for active and reactive power, which are from RES to VSC

3. Methodology –Projection Method



$\max p_{r2v}^{res}$

S.t. RES-Side Constraint

Key Steps of ECM

- Solve the max problem related to boundary active power
- Solve the max and min problem related to boundary reactive power
- Update the boundary active power with small reduction
- Stop until active power approach its minimum value

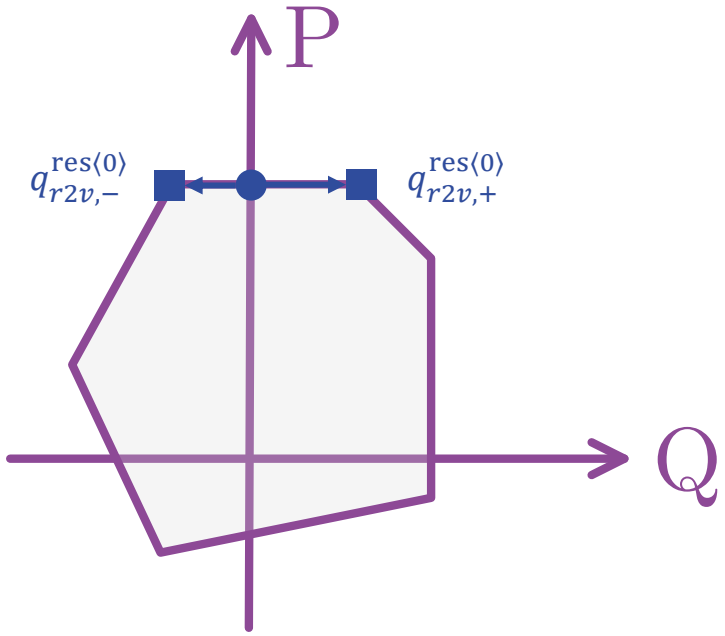
✎ ECM: Epsilon-Constrained Method

3. Methodology –Projection Method

$$\min/\max\ q_{r2v}^{res}$$

$$\hat{p}_{r2v}^{\mathrm{res}\langle n\rangle} - \epsilon \le p_{r2v}^{\mathrm{res}} \le \hat{p}_{r2v}^{\mathrm{res}\,\langle n\rangle} + \epsilon,$$

S.t. RES-Side Constraint

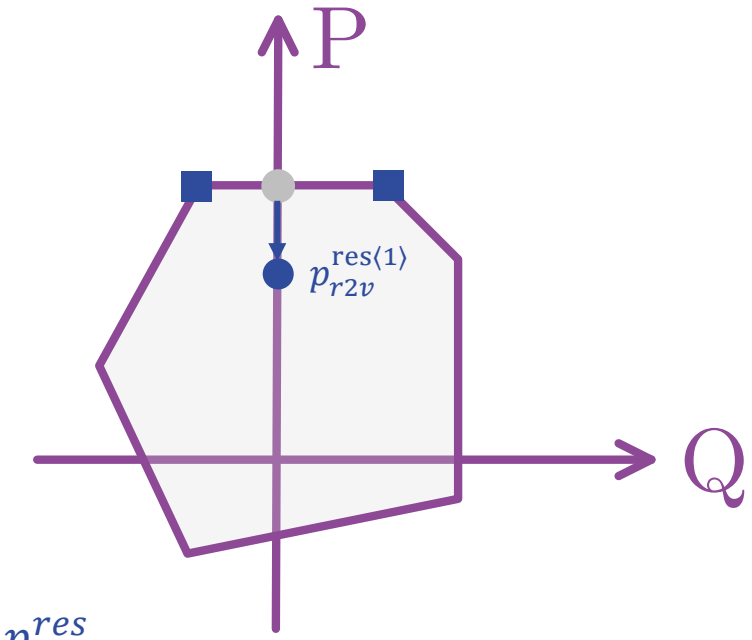


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3. Methodology –Projection Method



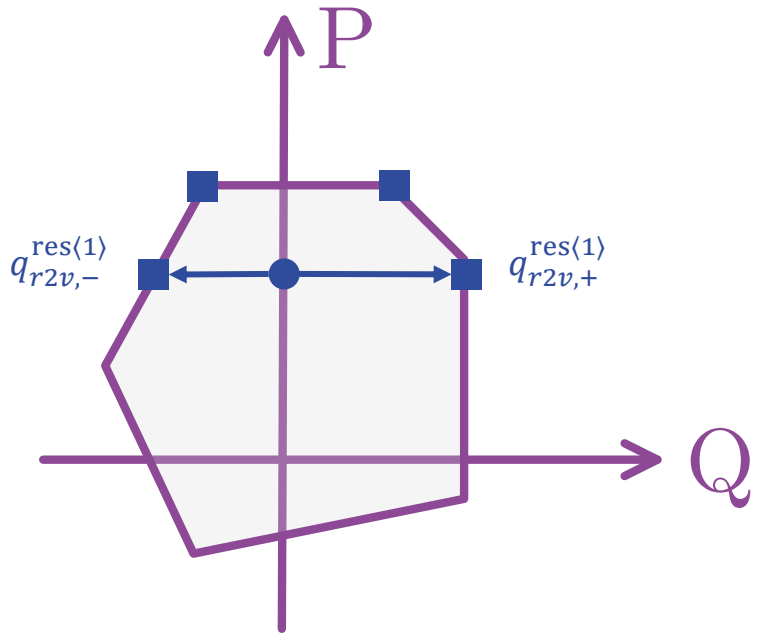
$$\hat{p}_{r2v}^{res\langle n+1\rangle} \leftarrow \hat{p}_{r2v}^{res\langle n\rangle} - \Delta p_{r2v}^{res}$$

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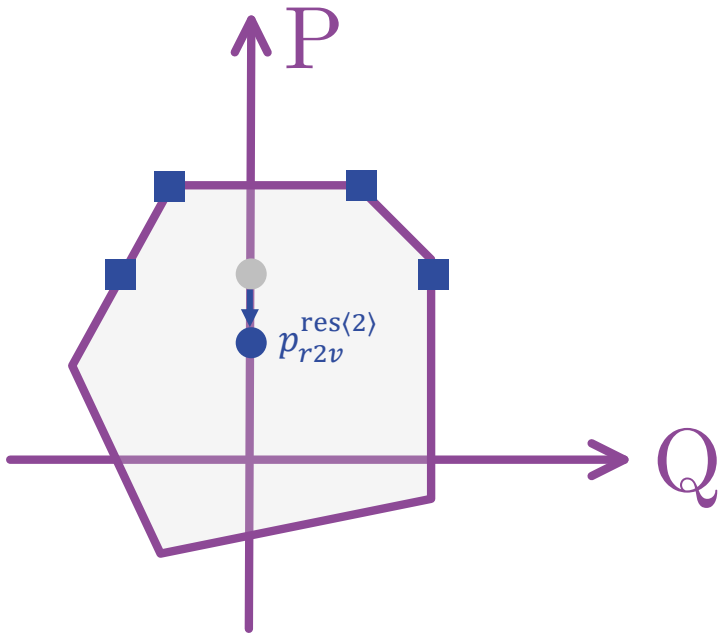


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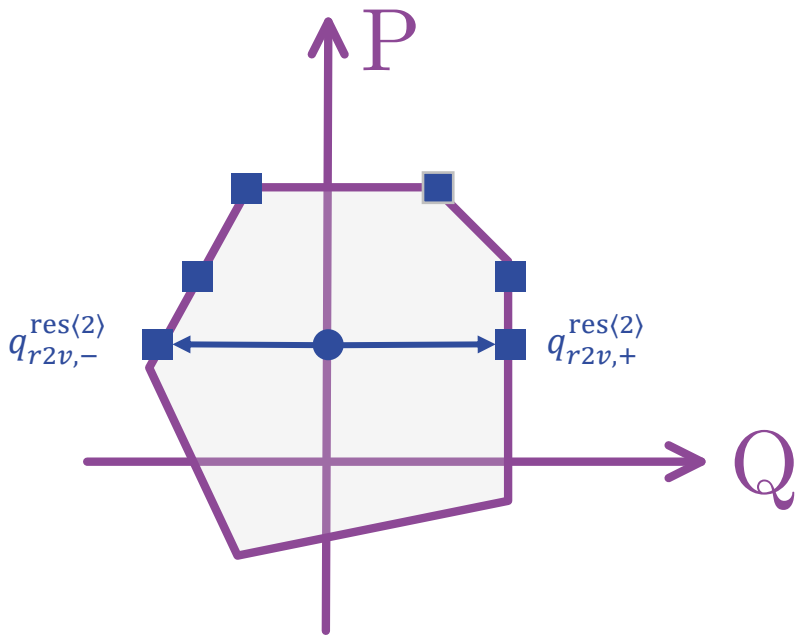


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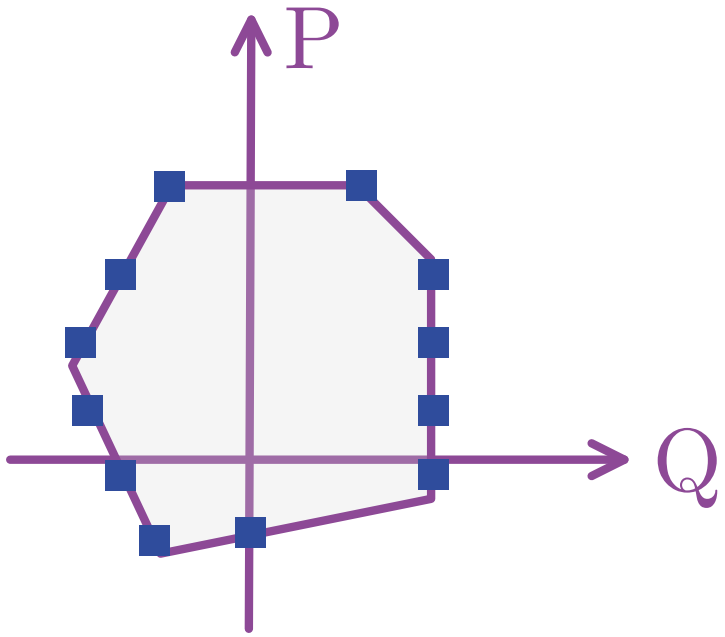


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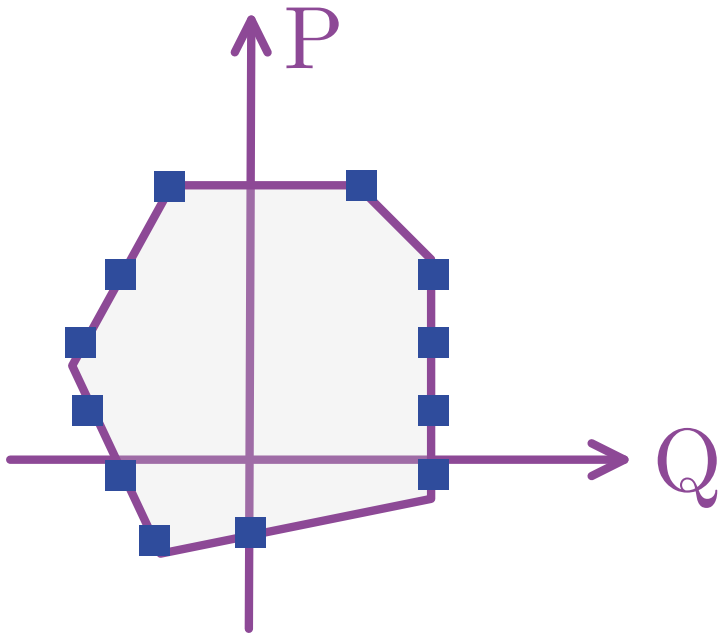


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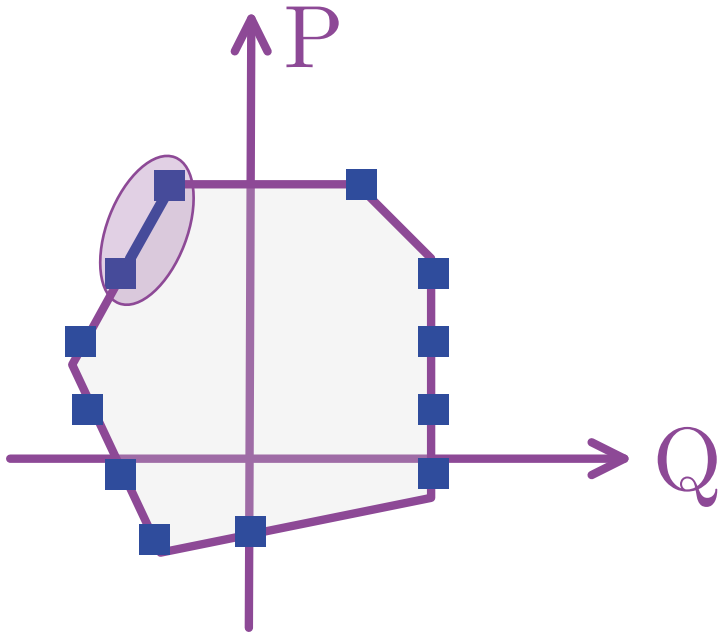
3. Methodology –Projection Method



$$\mathcal{P} := \bigcup_{n=1}^{N^{ecm}} \left\{ \left(\hat{p}_{r2v}^{\mathrm{res}\langle n \rangle}, \hat{q}_{r2v,-}^{\mathrm{res}\langle n \rangle} \right), \left(\hat{p}_{r2v}^{\mathrm{res}\langle n \rangle}, \hat{q}_{r2v,+}^{\mathrm{res}\langle n \rangle} \right) \right.$$

▶ Two-dimensional vertex set regrading boundary active/reactive power from RES to VSC is obtained

3. Methodology –Projection Method



$$a_n p_{r2v}^{res} + b_n q_{r2v}^{res} \le c_n \quad \begin{cases} a_n := q_{r2v}^{\mathrm{res}\langle n+1\rangle} - q_{r2v}^{res\langle n\rangle} \\ b_n := p_{r2v}^{res\langle n\rangle} - p_{r2v}^{res\langle u+1\rangle} \\ c_n := a_n p_{r2v}^{res\langle n\rangle} + b_n q_{r2v}^{res\langle n+1\rangle} \end{cases}$$

- For each edge formed by two adjacent vertices, a corresponding linear constraint can be formulated. The set of these linear constraints form the projected PQ feasible region

3. Methodology –Remarks

Communication Requirements: ADMM requires boundary-information exchange at every iteration until convergence, whereas ECM requires **only a one-shot** exchange of the RESs' P–Q feasible regions with the AC/DC grid operator.

Computation Burdens: The relative computational performance of ADMM and ECM varies depending on the number of iterations required for ADMM convergence and the resolution(step size for adjusting the active power) adopted in ECM.

Optimality: For the SOCP-based AC/DC OPF, ADMM can theoretically equal to the centralized optimum upon convergence. ECM lacks the same optimality guarantee because it uses an approximated RES P–Q region. Nevertheless, ECM can achieve comparable results when the approximation is sufficiently accurate and the optimal operating point lies close to the projected boundary.

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4. Case Studies –Case system

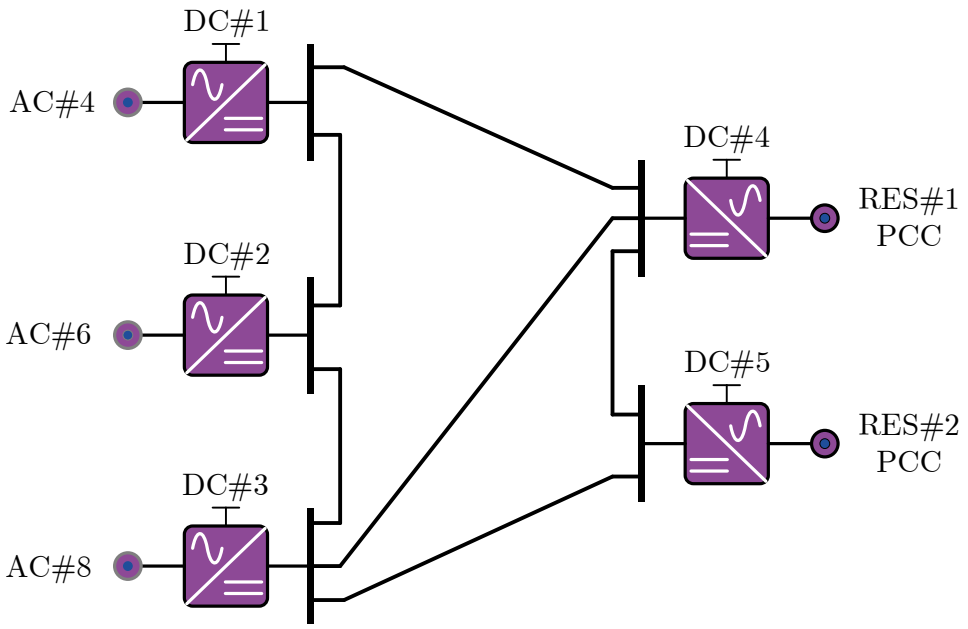


Fig: Toy case layout. DC side is a 5-terminal DC network. Terminal #1~#3 are respectively connected with AC bus #4, #6, #8 of a 9-bus AC grid . Terminal #4 and #5 are respectively connected with RES#1 and RES#2, which are both of 30-bus radial feeder .

More details and source code are available @

https://github.com/control-protection-grids-tudelft/AC-DC-OPF-RES-Privacy-Preservation

4. Case Studies –Why internal operational constraints of RESs need to be considered?

Table: Comparison of optimized generator real power with/without RES detailed constraints

AC/DC OPF Result	RES Internal Operational Cons. ✗	RES Internal Operational Cons. √	Relative Error [%]
Generation Cost [$ /h]	3683.047	4099.690	10.163
P_{G1}[MW]	10.000	10.000	0
P_{G2}[MW]	67.486	79.088	14.670
P_{G3}[MW]	125.941	133.808	5.879

Relative Error: Model considering RES internal operation constraints is the benchmark

RES Internal Operational Cons include:

- Lindistflow power constraint
- Power injection balance
- Voltage limitation
- Power output constraint of the single RES unit

4. Case Studies –Iterative process related to ADMM (distributed)

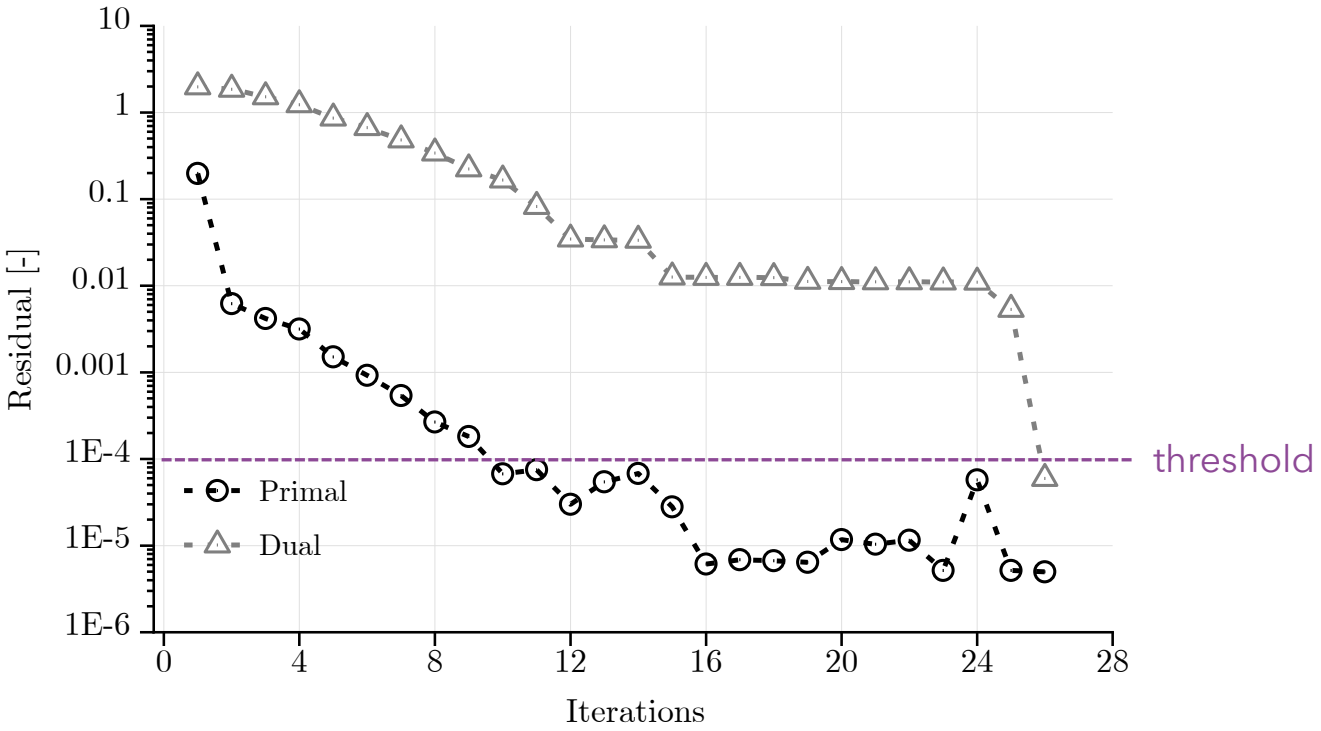


Fig: ADMM iteration process and residual convergence

▶ The residuals decrease gradually during the iterative process, and convergence is achieved after 26 iterations when both residuals satisfy the threshold.

4. Case Studies –PQ feasible region identified using ECM (projection)

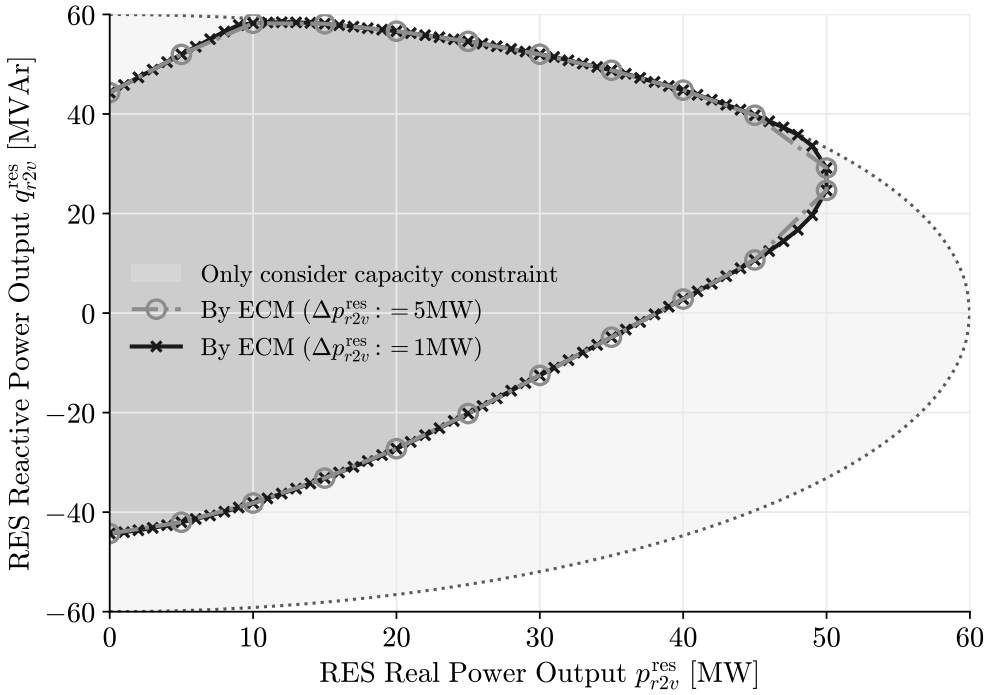


Fig: PQ region associated with RES boundary power

▶ ECM provides a more accurate PQ feasible region by accounting for RES internal operational constraints; finer sweeping resolution improves accuracy.

4. Case Studies –Comparison among centralized optimization, ADMM, and ECM

Table: Comparison of AC/DC OPF results among centralized optimization, ADMM, and ECM

AC/DC OPF Result	Centralized Optimization	ADMM	ECM
Generation Cost [$ /h]	4099.690	4100.064	4099.008
P_{G1}[MW]	10.000	10.000	10.000
P_{G2}[MW]	79.088	79.070	79.068
P_{G3}[MW]	133.808	133.827	133.796
Solver Time [s]	0.0767	2.1169	1.8601

- The YALMIP's built-in timer is used for measuring the computation time. Warm-up run was firstly conducted to mitigate the impact of initialization effects. Then the problem was solved five additional times, and here the average time is reported.

▶ Both ADMM and ECM exhibit decent optimization performances, and the calculated generator outputs are very close to the benchmark(centralized optimization).

5. Key Takeaways

- Numerical experiments demonstrate that Distributed (ADMM) and Projection (ECM) methods both are promising to handle RES-side privacy preserving issue and can achieve solutions **close to** the centralized optimization benchmark ʕ̢̅/'˳˳'\

- For the considered test case, ECM outperforms ADMM in terms of **execution time**. However, this does not imply that the Projection method is universally superior to Distributed method, as system structures and parameter settings may affect the performance of both methods (⊙_⊙)!

- Future work will further account for **uncertainty** in RES and investigate how to extend the Distributed (ADMM) and Projection (ECM) methods to optimization under uncertain operating conditions (˘̀_˘́)ʊ

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Thank You for Your Attention (ㅅ´˘`)✧